Architecting Agentic Software Systems

A Technical Portfolio and Professional Itinerary

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Abstract

This document is a technical overview of my work designing, building, and deploying *agentic software systems*: autonomous and semi-autonomous programs that plan, act, and verify their own work under engineered constraints. I am the Chief Technology Officer of Finsider, where I built and deployed Symphony, a supervised-autonomy software-delivery runner that drives product tickets to reviewed, evidence-backed pull requests. Before this I studied Computer Science at Queen's University, then raised $220,000 at a $2.2M valuation as co-founder and CEO of an Antler Canada–backed company at the age of 19, and authored single-author research formalizing the reliability of multi-step language-model reasoning, validated against four frontier models to within roughly one percent of empirical accuracy. A second single-author paper, *The Information-Maintenance Hypothesis*, argues that aging, intelligence, and markets are the same problem in information theory. The unifying thread of this portfolio is a specific, defensible engineering thesis about autonomous AI, derived from that research and realized in production. This document doubles as a professional itinerary; my curriculum vitae is appended. Every system referenced is hyperlinked to its public source repository.

1 Introduction

I build artificial-intelligence systems for the highest-stakes work in finance, and I build the autonomous infrastructure that builds them. My work spans three layers that most engineers treat separately: the *theory* of why language-model systems fail, the *systems* that exploit that theory in production, and the *tooling* that lets a single engineer operate at the throughput of a team.

The defining specialization is *agentic software*: programs that decompose a goal into steps, take real actions in the world (writing code, querying corpora, operating a browser, transacting), and judge the quality of their own output. Agentic systems are powerful and notoriously unreliable; the central engineering problem is making them trustworthy enough to point at production. The systems in this portfolio are my answers to that problem, and Section 3 gives the formal reason they are designed the way they are.

Selected facts. Chief Technology Officer of Finsider on a $200,000 salary at the age of 20; raised $220,000 at a $2.2M valuation (Antler Canada Fund I SAFE) as co-founder/CEO before turning 20; single-author research validated against GPT-5, Claude Opus 4.6, Claude Sonnet 4.6, and Gemini 3.1 Pro; author of SYMPHONY, deployed live in a commercial engineering environment.

2 Core Technical Competencies

Agentic systems (primary specialization). Supervised-autonomy architectures; ticket-native state machines; acceptance-contract prompting; evidence-gated delivery pipelines; multi-agent orchestration with shared context; tool-use design; process supervision, watchdogs, and safe automated Git; failure recovery with bounded retry.

Applied AI / LLMs. LLM orchestration; Retrieval-Augmented Generation (Vertex AI RAG; corpus-per-workload; parallel multi-query retrieval with dedup/ranking); structured extraction with calibrated confidence; process reward models; frontier-model evaluation; Anthropic, Google Vertex AI, and Gemini 2.0 Flash APIs.

Languages. Python, TypeScript/JavaScript, Swift (iOS), SQL, Bash, LaTeX.

Full-stack. Next.js 16, React 19, Tailwind v4, shadcn/ui; Supabase (PostgreSQL, Row-Level Security, Auth, Storage), Firebase, Sanity (headless CMS); Stripe, Resend, Twilio; server-sent-event streaming, async job queues, webhook pipelines.

Cloud & infrastructure. Google Cloud Platform (Compute Engine, Cloud Storage, Vertex AI), Vercel; self-hosted Proxmox cluster, Docker, `systemd`, macOS `launchd`, Tailscale; CI-aware multi-repository delivery.

Applied mathematics. Probability theory; Bernoulli and Hoeffding bounds; maximum-likelihood estimation; the delta method and confidence-interval construction; empirical model evaluation.

Financial domains. M&A Quality of Earnings; Canadian mortgage underwriting (GDS/TDS, OSFI guidelines, stress testing); financial-document AI; document-authenticity and fraud-signal detection.

3 Research Foundations: The Reliability of Agentic Reasoning

The design of every agentic system I build follows from a result I proved in *Uncertainty Propagation in Tree-Structured Language Model Reasoning* (github.com/dm3n/uncertainty-propagation). Model each step of an autonomous process as a Bernoulli trial with step-specific failure probability θ_i.

Theorem 1 (Chain success)**.** *For an n-step reasoning chain $\mathcal{C}_n$ with conditionally independent steps,*

$$\mathbb{P}(\textit{CORRECT} \mid \mathcal{C}_n, \boldsymbol{\theta}) \;=\; \prod_{i=1}^{n}(1-\theta_i).$$

Corollary 1 (Exponential decay)**.** *With mean failure rate $\bar{\theta}$, $\mathbb{P}(\textit{CORRECT}) \leq (1-\bar{\theta})^n \leq e^{-n\bar{\theta}}$.*

The practical consequence is severe: a long, unsupervised agent run decays exponentially in reliability, so it *cannot* be trusted to ship. The paper further proves that a k-ary, depth-d *tree* with majority aggregation decays only sub-exponentially in total compute, $\exp(-\Omega(N^{1-1/(d+1)}))$, and derives the compute-optimal branching factor and depth, together with maximum-likelihood estimators and delta-method confidence intervals for end-to-end success. Empirically, the model predicts observed accuracy to within $\sim$1% across GPT-5, Claude Opus 4.6, Claude Sonnet 4.6, and Gemini 3.1 Pro; at a fixed 48-step budget a chain succeeds $\approx 0.2\%$ of the time while a depth-3 tree succeeds 97.9%.

This yields the engineering principle that organizes the rest of this portfolio.

Principle 1 (Supervised, tree-structured autonomy)**.** Do not ship from one long autonomous chain. Decompose work into short, independently verified units; aggregate across them with explicit checkpoints (validation, evidence, human review). Keep per-step failure low and let aggregation, not chain length, carry reliability.

4 Symphony: A Supervised-Autonomy Delivery System

Symphony (github.com/dm3n/symphony) is the clearest demonstration of my agentic-software engineering. It is a Jira-native, evidence-gated runner that takes a product ticket from backlog to a reviewed, evidence-backed pull request, with a human as the sole approval authority. I designed and built it as CTO of Finsider, where it runs live against a multi-repository codebase. It is a ∼2,300-line, standard-library Python orchestrator (no agent framework) chosen for auditability and portability.

Architecture. The system treats the coding agent as a capable but unprivileged developer: it may implement, validate, and produce evidence, but never holds approval authority. Its defining properties:

- **Jira as the state machine.** Ticket status and labels *are* the control flow; there is no external queue or database. Because state is re-derived from Jira on every poll, the daemon is crash-safe and losslessly resumable, a direct application of the "keep each step verifiable" principle.
- **Per-ticket isolation.** Each ticket runs in an isolated Git workspace on its own branch, with pinned commit identity and evidence excluded from history.
- **Acceptance contracts.** Acceptance criteria are parsed from the ticket and injected into the agent prompt as an explicit verification checklist.
- **Evidence gating.** No ticket reaches review, and no pull request is created, without Playwright-captured screenshots/video of the running result, integrity- checked against minimum-size thresholds to reject empty or synthetic "evidence."
- **Intent-classified human review.** Reviewer comments are classified (question vs. rework vs. approval vs. neutral), so the loop answers questions in place and re-runs the agent only on genuine change requests.
- **Cross-repo scope critic.** The system detects when a ticket requires changes across multiple repositories and blocks review if coverage is incomplete.

- **Safety by construction.** `--force-with-lease` only; recursive process-tree cleanup and a dev-server watchdog; bounded failure auto-retry with cooldown; a single-instance file lock.
- **Two deployment targets.** macOS `launchd` for local operation and an always-on GCP Compute Engine VM under `systemd` for production.

The invariant. The controls compose to a single guarantee: SYMPHONY can never merge code, never force-clobber a branch, never advance unverified work, and never fail silently. The maximum blast radius of any malfunction is a clearly labeled, un-merged draft pull request plus a notification, exactly the property required of an autonomous system pointed at production. The full engineering record (architecture, state machine, hardening, deployment, security, and sanitized reference interfaces) is published at `github.com/dm3n/symphony`.

5 A Portfolio of Agentic and AI Systems

Airbank: Quality-of-Earnings engine (`github.com/dm3n/airbank`). An eleven-section, RAG-grounded extraction pipeline that produces analyst-grade M&A Quality-of-Earnings workbooks in which every figure is traceable to a source document. It applies Principle 1 to document AI: rather than one long extraction pass, each section is an independently verified unit with calibrated confidence, audit-grade citation (document, page, verbatim excerpt), and a policy of flagging missing data rather than fabricating it; a deterministic path bypasses retrieval for clean spreadsheets, and a formula resolver reconciles cross-section dependencies over a live SSE stream.

Airbank: mortgage underwriting (`github.com/dm3n/airbank-mortgage-platform`). Full-lifecycle AI origination for Canadian brokers: Canadian-specific document extraction (T4, NOA, pay-stub, LOE); *fraud-signal detection* via PDF-metadata, font-consistency, and name/income cross-checks; a GDS/TDS engine with stress-test rate; one-call intake orchestration; real-time upload verification; and an autonomous, cron-driven pipeline agent that generates role-aware priorities across a live deal pipeline.

Macintosh: a multi-agent engineering operating system (`github.com/dm3n/macintosh`). A reproducible, self-hosted environment that lets one engineer ship like a team: a six-stage agentic coding pipeline expressed as Claude Code skills;

four coding agents (Claude Code, Codex, Gemini, OpenCode) sharing one context; a Personal Knowledge Brain (Obsidian + an LLM compiler) for persistent memory; and a Proxmox/Docker homelab with a human-gated approval model.

VisionClaw: a real-time multimodal agent (github.com/dm3n/VisionClaw). An accessibility-first iOS agent (Swift/SwiftUI, Meta Wearables DAT SDK, Gemini Live over WebSocket) that streams video from Meta Ray-Ban glasses and proactively narrates the world for visually impaired users, with hands-free activation and resilient session recovery.

The Information-Maintenance Hypothesis: a second research paper (github.com/dm3n/information-maintenance-hypothesis). A single-author theory paper arguing that three problems, why organisms age, what intelligence is, and how markets price the world, are one problem: maintaining a low-entropy predictive model against entropy at a free-energy cost. It is anchored on two results that are theorems rather than analogies, Landauer's principle (maintaining a bit of information costs at least $k_{\mathrm{B}}T \ln 2$ of energy) and the Kelly–Cover identity (capital growth rate equals mutual information with the future), and it closes with eleven falsifiable predictions across biology, AI, and finance. Written in LaTeX with four figures and three appendices, it demonstrates the same first-principles modeling, from formal definition to proven result, that underwrites the engineering thesis above.

Further systems. **ROGI** (github.com/dm3n/rogi), an AI-native mortgage-intake funnel with a Canadian qualification engine (first version sold for $7,500); **Sacred Secretion Agent** (github.com/dm3n/sacred-secretion-agent), a fully autonomous scheduling/email agent; **Nassau** (github.com/dm3n/nassau), a complete e-commerce platform built from scratch (Next.js, Sanity CMS, Stripe, Firebase); and a second research manuscript (github.com/dm3n/human-divinity).

6 Itinerary: Roles, Milestones, and Technical Objectives

Primary location of work: Toronto, Ontario, Canada (Antler Canada / OneEleven, 325 Front St W).

6.1 Co-founder & CEO: Flagpost AI Inc. (Sep 2025 – Q1 2026)

- Selected into the Antler Canada TOR8 residency (commenced Sep 2025) and passed the Antler Investment Committee.
- *Funding milestone:* raised $220,000 at a $2.2M valuation via a SAFE from Antler Canada Fund I, L.P.; incorporated Flagpost AI Inc. (Nov 2025).
- *Duties:* company formation, team and cap-table structure, product direction (AI-native sell-side M&A preparation), and investor relations.
- *Outcome:* stepped aside to pursue the broader M&A-intelligence thesis.

6.2 Founder: Airbank (Q1 2026)

- *Technical objectives:* design an AI-native operating system for M&A, beginning with the Quality-of-Earnings wedge; build the QoE and mortgage platforms above.
- *Development phases:* RAG ingestion → section-wise extraction with confidence flags → auditable source-linked cells → reconciliation/export; mortgage intake → AI verification → underwriting audit → pipeline orchestration.
- *Customer acquisition:* signed LOIs, NDAs, and advisory agreements with senior professionals across private credit, M&A advisory, investment banking, and private equity.

6.3 Chief Technology Officer: Finsider (2026 – Present)

- *Compensation:* $200,000 annual salary, materially above the norm for the role and tenure.
- *Technical objectives:* automate Quality of Earnings end-to-end; own the multi-repository platform (frontend, backend engine, AI/agent services) and the financial-correctness layer (Balance Sheet, P&L, Cash Flow, and Cash Proof tie-out with standardized export).
- *Signature deliverable:* designed and deployed Symphony as the company's agentic delivery infrastructure.
- *Forward plan (petition period):* (i) harden and scale the QoE automation and agentic delivery pipeline; (ii) extend financial-statement correctness

and export coverage; (iii) grow enterprise adoption of the automated QoE product; (iv) advance the supervised-autonomy research-to-production loop.

7 Education

Queen's University, Computer Science. Left to build full-time after raising venture funding.

Repository Index

- **Portfolio (this overview):** `github.com/dm3n/portfolio`
- **Symphony:** `github.com/dm3n/symphony`
- **Macintosh (engineering OS):** `github.com/dm3n/macintosh`
- **Research, reasoning reliability:** `github.com/dm3n/uncertainty-propagation`
- **Research, information-maintenance hypothesis:** `github.com/dm3n/information-maintenanc`
- **Research, manuscript:** `github.com/dm3n/human-divinity`
- **Airbank QoE** (private): `github.com/dm3n/airbank`
- **Airbank Mortgage** (private): `github.com/dm3n/airbank-mortgage-platform`
- **ROGI:** `github.com/dm3n/rogi` | `github.com/dm3n/rogi-v2`
- **VisionClaw:** `github.com/dm3n/VisionClaw`
- **Nassau:** `github.com/dm3n/nassau`
- **Sacred Secretion Agent:** `github.com/dm3n/sacred-secretion-agent`
- **Other:** `github.com/dm3n/nodebase` | `github.com/dm3n/ratestore` | `github.com/dm3n/OpenCV`

Prepared June 27, 2026. Curriculum vitae appended. Sensitive investor, customer, and financial specifics are maintained in a private data room available on request.